

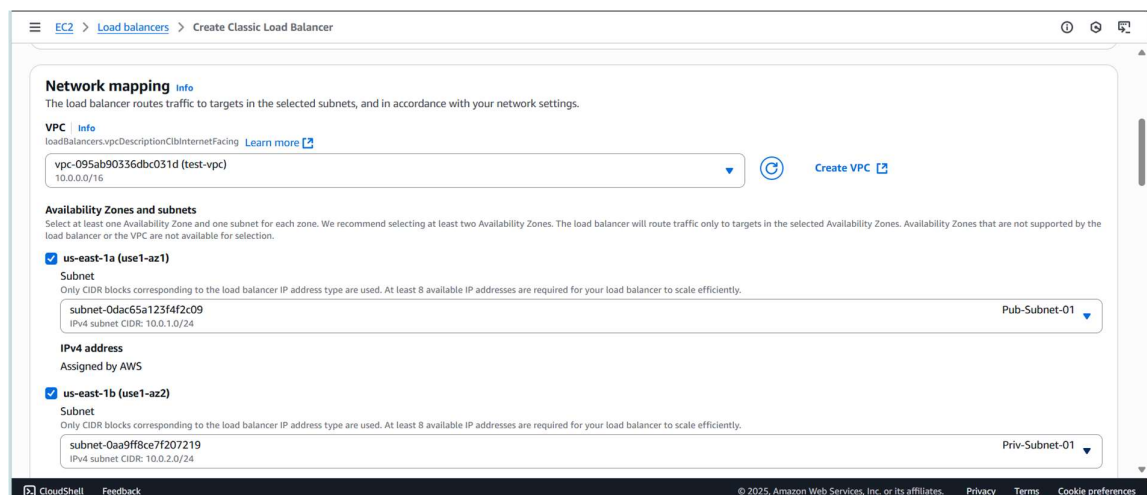
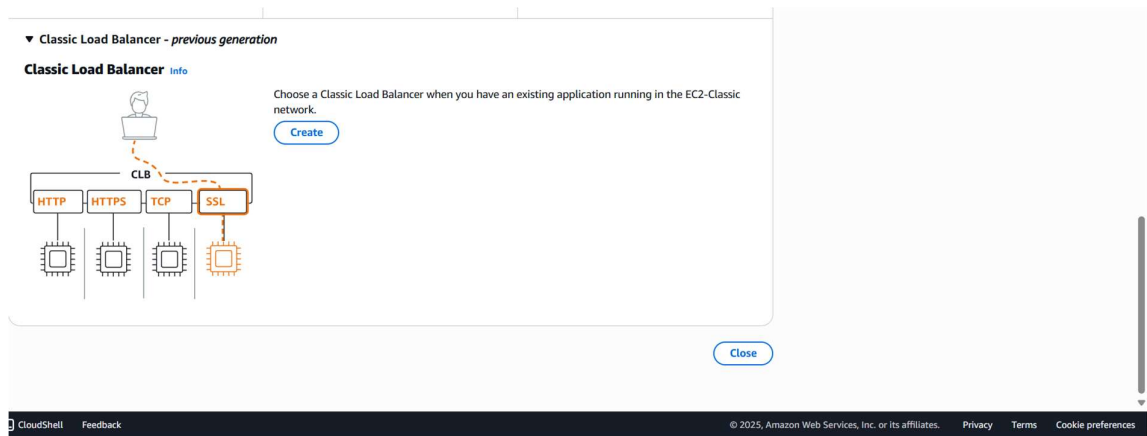
Task on Load Balancers

1. Configure a Classic Load Balancer.

Purpose: A Classic Load Balancer in AWS is designed to distribute incoming application or network traffic across multiple Amazon EC2 instances, ensuring no single instance bears too much load. By operating across different Availability Zones, it enhances the availability and fault tolerance of applications. It supports both HTTP/HTTPS and TCP/SSL traffic, performs regular health checks on instances, and routes traffic only to healthy ones.

Create Load Balancer:

- Go to Load Balancers → “Create Load Balancer” → Classic Load Balancer



- Listeners: HTTP on port 80
- VPC: MyVPC
- Availability Zones: Select subnets (public and private)
- Security group: Allow inbound HTTP (port 80) from anywhere
- Target group: Select MyTargetGroup
- Click Create

EC2 > Load balancers > Create Classic Load Balancer

Network mapping Info

The load balancer routes traffic to targets in the selected subnets, and in accordance with your network settings.

VPC Info
[loadbalancers.vpcDescriptionCblnternetfacing](#) [Learn more](#)

vpc-095ab90336d6c031d (test-vpc)
 10.0.0.0/16

[Create VPC](#)

Availability Zones and subnets
 Select at least one Availability Zone and one subnet for each zone. We recommend selecting at least two Availability Zones. The load balancer will route traffic only to targets in the selected Availability Zones. Availability Zones that are not supported by the load balancer or the VPC are not available for selection.

☒ **us-east-1a (use1-az1)**
 Subnet
 Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
 subnet-0dac65a123f4f2c09
 IPv4 subnet CIDR: 10.0.1.0/24 Pub-Subnet-01

IPv4 address
 Assigned by AWS

☒ **us-east-1b (use1-az2)**
 Subnet
 Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
 subnet-0aa9ff8ce7f207219
 IPv4 subnet CIDR: 10.0.2.0/24 Priv-Subnet-01

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EC2 > Instances > Launch an instance

Select

User data - optional Info
 Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
yum install nginx -y
systemctl start nginx

echo "<html><head><title>Server-1</title></head><body><h1>server1</h1></body></html>" > /usr/share/nginx/html/index.html

systemctl restart nginx
```

☐ User data has already been base64 encoded

Summary

Number of instances Info
 1

Software Image (AMI)
 Amazon Linux 2023 AMI 2023.9.2...[read more](#)
 ami-052064a798f08f0d3

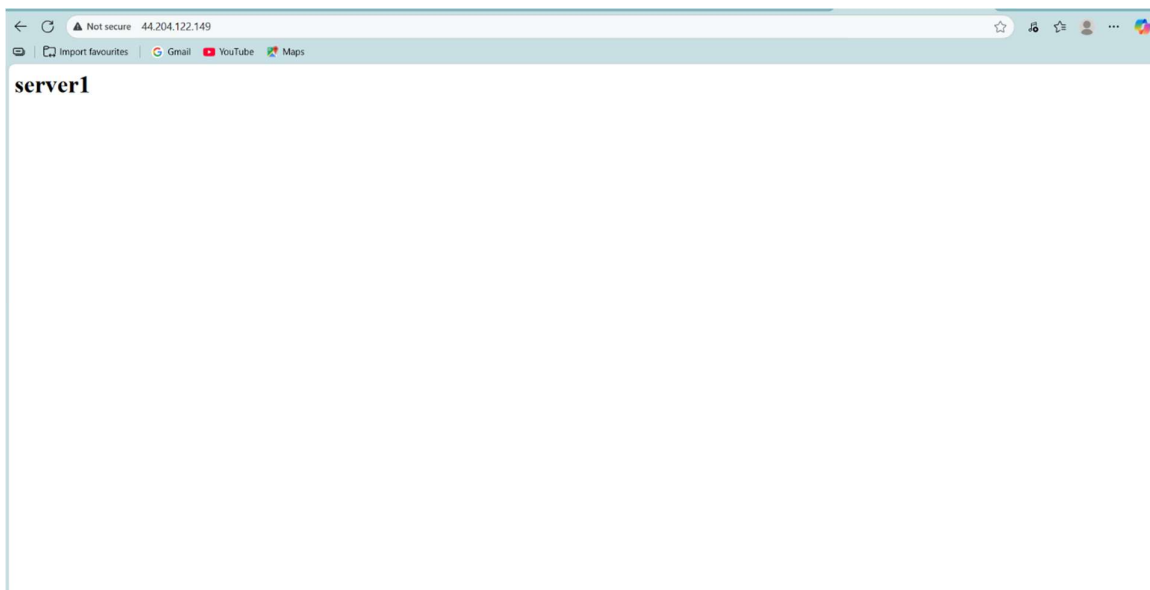
Virtual server type (instance type)
 t3.micro

Firewall (security group)
 New security group

Storage (volumes)
 1 volume(s) - 8 GiB

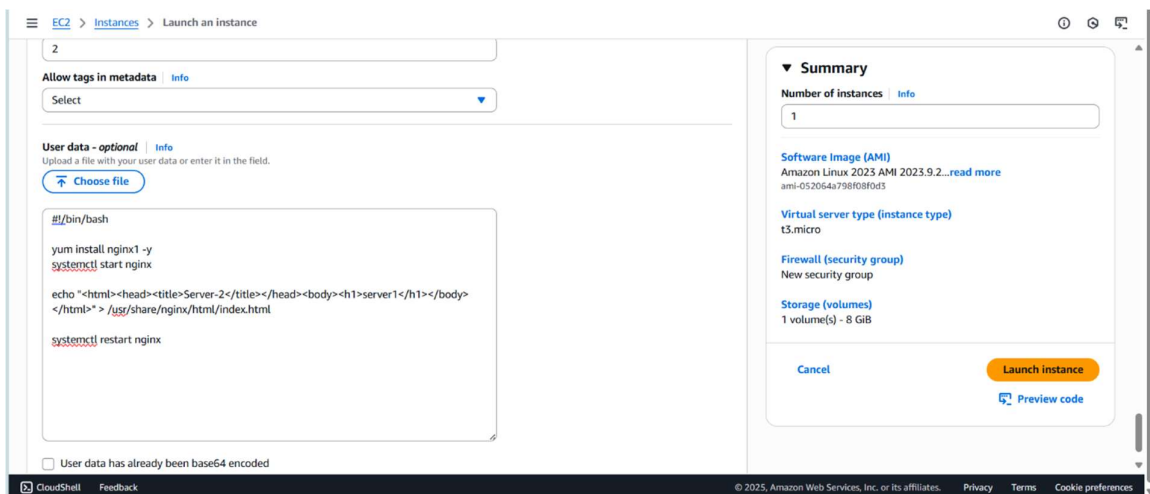
[Cancel](#) [Launch instance](#) [Preview code](#)

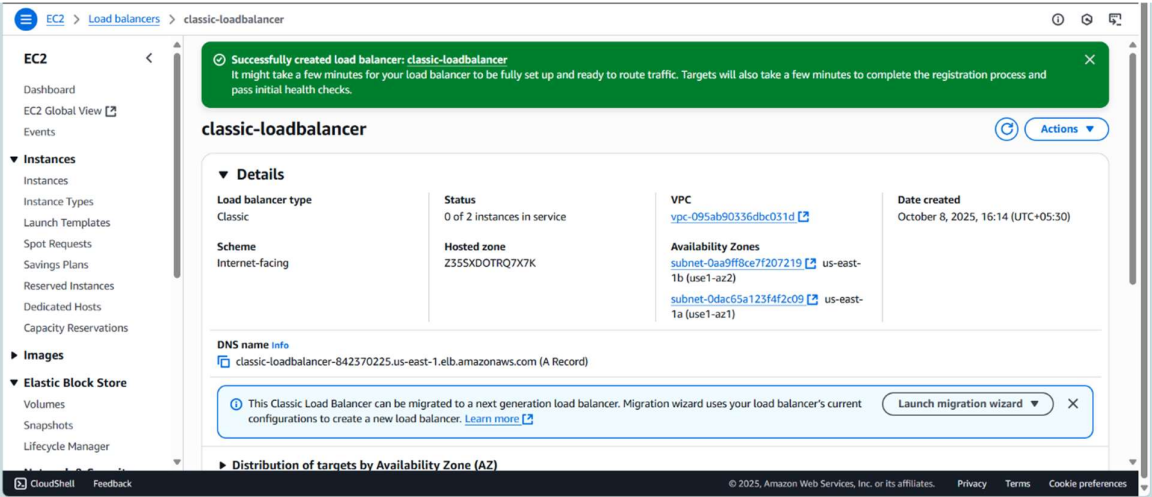
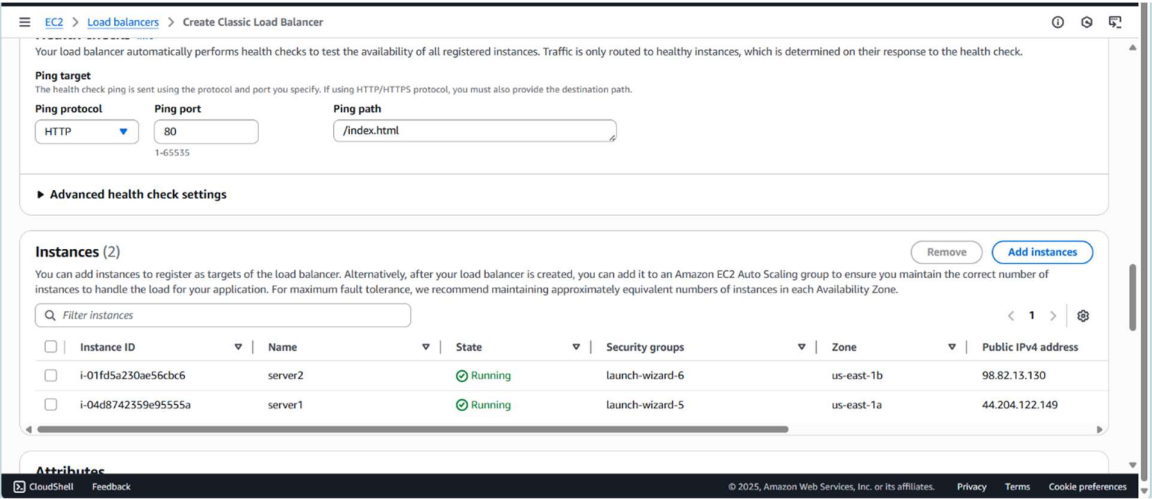
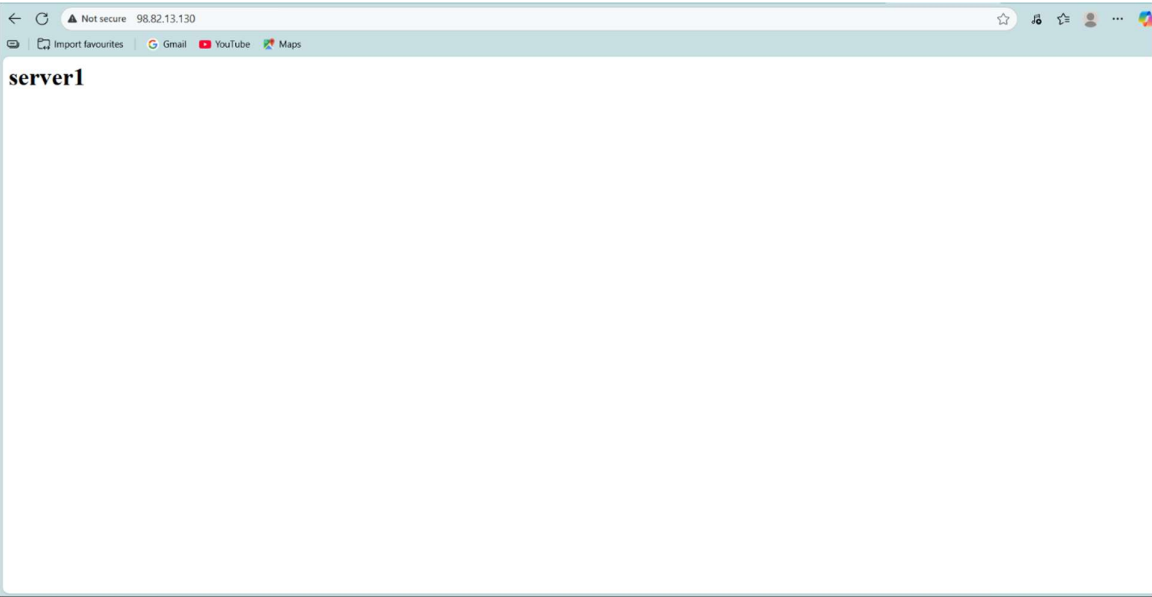
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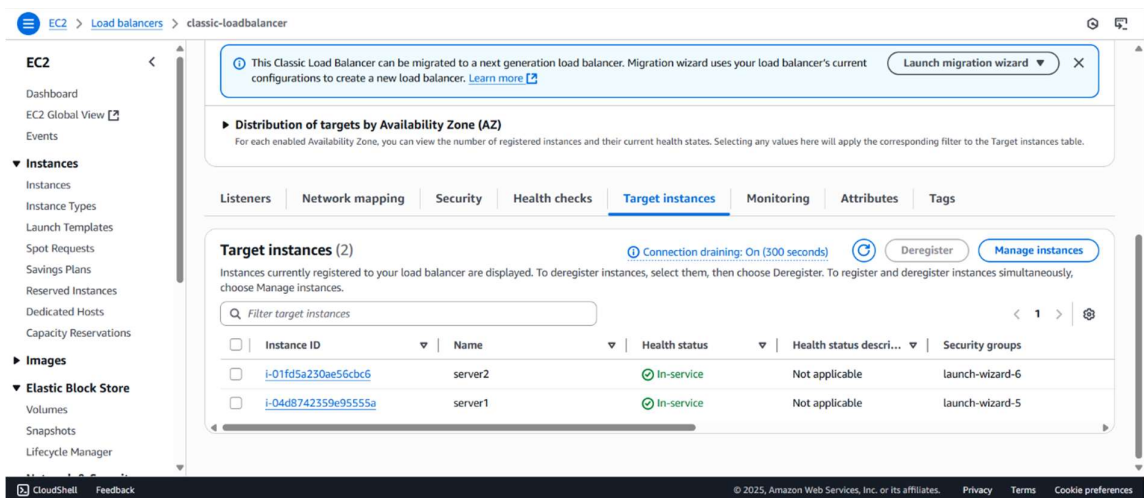
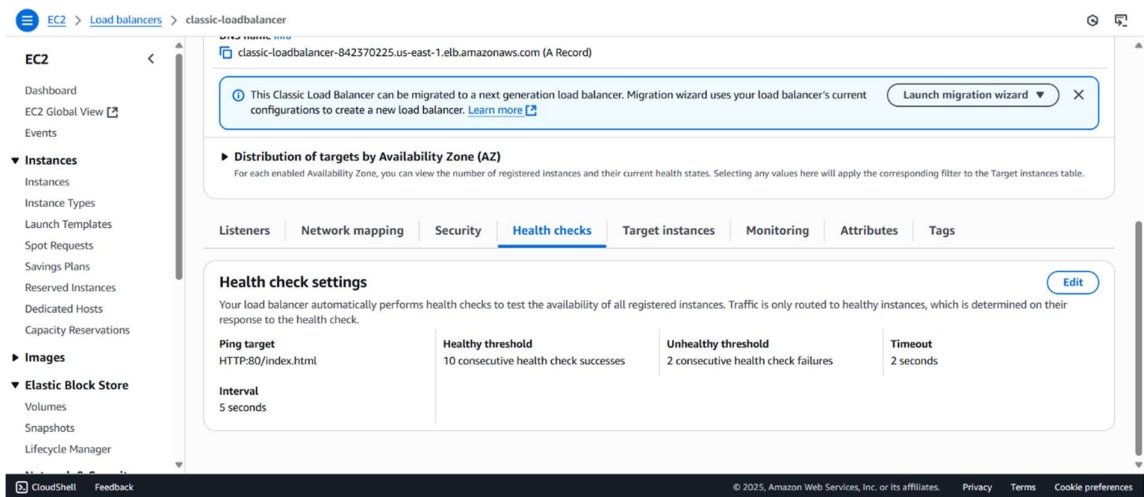


```
[ec2-user@ip-10-0-1-26 ~]$ systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; p
   Active: active (running) since Wed 2025-10-08 10:32:24 UTC; 19s ago
   Process: 3045 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exit>
   Process: 3103 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=>
   Process: 3148 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCC>
   Main PID: 3160 (nginx)
   Tasks: 3 (limit: 1053)
   Memory: 3.2M
   CPU: 69ms
   CGroup: /system.slice/nginx.service
           └─3160 "nginx: master process /usr/sbin/nginx"
             └─3161 "nginx: worker process"
               └─3162 "nginx: worker process"

Oct 08 10:32:24 ip-10-0-1-26.ec2.internal systemd[1]: Starting nginx.se>
Oct 08 10:32:24 ip-10-0-1-26.ec2.internal nginx[3103]: nginx: the confi>
Oct 08 10:32:24 ip-10-0-1-26.ec2.internal nginx[3103]: nginx: configura>
Oct 08 10:32:24 ip-10-0-1-26.ec2.internal systemd[1]: Started nginx.ser>
lines 1-19/19 (END)
```

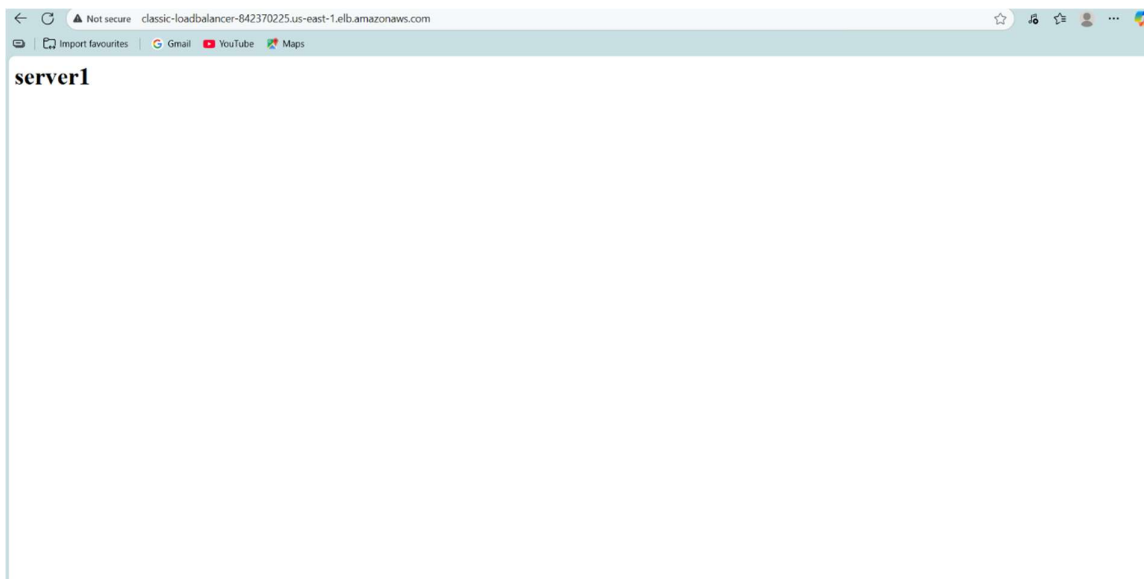






Test:

- Get the DNS name of the load balancer
- Open it in a browser → You should see the server details from either instance

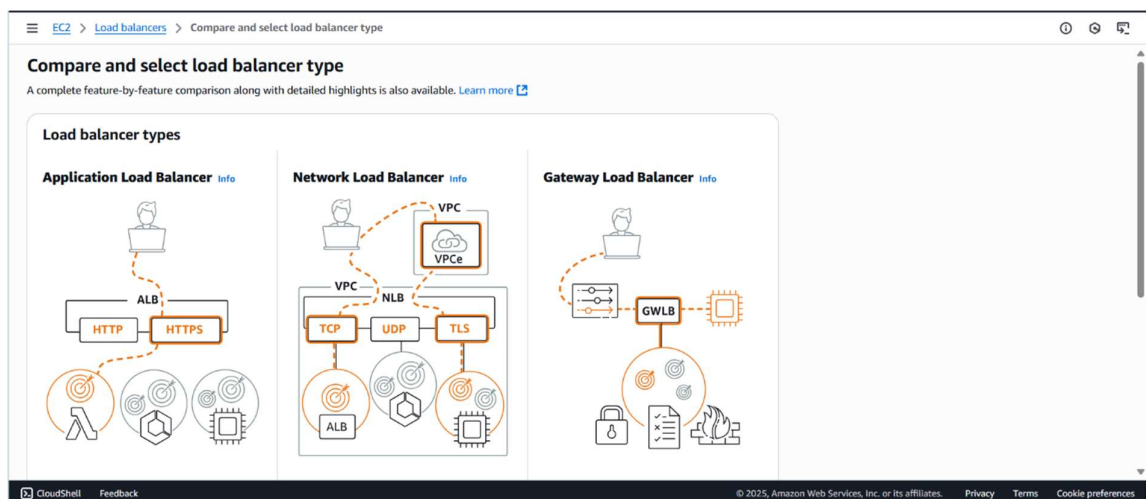


2. Configure the application load balancer.

Purpose: An **Application Load Balancer (ALB)** is a sophisticated AWS service designed to distribute incoming **HTTP/HTTPS** traffic across multiple targets, such as Amazon EC2 instances, containers, and IP addresses, based on advanced routing rules. Unlike the Classic Load Balancer, the ALB operates at the **application layer (Layer 7)** of the OSI model, enabling it to make intelligent routing decisions based on content type, host, path, or even custom headers. The ALB is particularly useful for modern applications requiring **sophisticated traffic management**, such as web applications, APIs, and microservices architectures.

Steps to Create a Load Balancer:

- Go to Load Balancers → “Create Load Balancer” → Application Load Balancer



- Name: us-loadbalancer
- Scheme: internet-facing
- IP type: IPv4

EC2 > Load balancers > Create Application Load Balancer

How Application Load Balancers work

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.

US-Load balancer

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type [Info](#)
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**
Includes only IPv4 addresses.

☐ **Dualstack**
Includes IPv4 and IPv6 addresses.

☐ **Dualstack without public IPv4**
Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.

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- Listeners: HTTP on port 80
- VPC: MyVPC
- Availability Zones: Select subnets (public and private)
- Security group: Allow inbound HTTP (port 80) from anywhere
- Target group: Select MyTargetGroup
- Click Create

Create Target Group:

- Go to EC2 Dashboard → Target Groups → “Create target group”
- Type: Instances
- Name: us-east

EC2 > Target groups > Create target group

• Offers the flexibility for a Network Load Balancer to accept and route TCP requests within a specific VPC.
• Facilitates using static IP addresses and PrivateLink with an Application Load Balancer.

Target group name

US-East

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Protocol
Protocol for load balancer-to-target communication. Can't be modified after creation.

HTTP

Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

IP address type
Only targets with the indicated IP address type can be registered to this target group.

☒ **IPv4**
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ **IPv6**
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-0b6f0bdcee88ebc07 (us-east-01-vpc)

192.168.0.0/16

[Create VPC](#)

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- Protocol: HTTP, Port: 80
- VPC: MyVPC
- Health checks: Path /
- Register both EC2 instances
- Click Create

EC2 > Target groups > Create target group

Step 1: Create target group
Step 2: **Register targets**

Register targets

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2/2)

Filter instances

☒	Instance ID	Name	State	Security groups	Zone
☒	i-0f4185ddc8636095d	pri-server	Running	launch-wizard-2	us-east-1a
☒	i-023a8b38a62e49c21	PUB-server	Running	launch-wizard-2	us-east-1a

2 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

[Include as pending below](#)

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be5cc28a19b

Listeners and rules (1) Info

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

Filter listeners

☐	Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL/TLS
☐	HTTP:80	- Forward to target group - ELB: 1 (100%) - Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicable

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EC2 > Security Groups > Create security group

Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name Info
ALP-SG
Name cannot be edited after creation.

Description Info
Allows SSH access to developers

VPC Info
vpc-095ab90336dbc031d (test-vpc)

Inbound rules Info

Type	Protocol	Port range	Source	Description - optional
HTTP	TCP	80	Custom	

[Add rule](#) [Delete](#)

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EC2 > Security Groups > sg-0eab75c213f997487 - ALP-SG

Security group (sg-0eab75c213f997487 | ALP-SG) was created successfully

sg-0eab75c213f997487 - ALP-SG

Details

Security group name ALP-SG	Security group ID sg-0eab75c213f997487	Description security group for loadbalancer	VPC ID vpc-095ab90336dbc031d
Owner 085794372055	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules (1)

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-0fa62dad79b469d06	IPv4	HTTP	TCP	80

EC2 > Target groups > Apache-TG

Apache-TG

Details

arn:aws:elasticloadbalancing:us-east-1:085794372055:targetgroup/Apache-TG:ad26e56fa892a37c

Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1	VPC vpc-095ab90336dbc031d
IP address type IPv4	Load balancer ELB		

2 Total targets

2 Healthy

0 Anomalous

0 Unhealthy

0 Unused

0 Initial

0 Draining

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets | Monitoring | Health checks | Attributes | Tags

Registered targets (2)

Anomaly mitigation: Not applicable

Deregister | Register targets

Test:

- Get the DNS name of the ALB (e.g., MyALB-NVirenia-123456.us-east-1.elb.amazonaws.com)
- Open it in a browser → You should see the server details from either instance

EC2 > Load balancers

Load balancers (1)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

Name	Availability Zones	Security groups	ARN	Date created
ELB	2 Availability Zones	2 Security groups	ELB-499350098.us-east-1.e...	October 7, 2025, 18:16 (U

0 load balancers selected

Select a load balancer above.

←↻⚠ Not secure elb-499350098.us-east-1.elb.amazonaws.com

🔍 Import favourites | 📧 Gmail | 📺 YouTube | 🗺 Maps

☆🔊🔖👤⋮

Server Details

Hostname: ip-10-0-2-203.ec2.internal

IP Address: fe80::1077:a9ff:fe4e:61cd%ens5

←↻⚠ Not secure elb-499350098.us-east-1.elb.amazonaws.com

🔍 Import favourites | 📧 Gmail | 📺 YouTube | 🗺 Maps

☆🔊🔖👤⋮

Server Details

Hostname: ip-10-0-1-76.ec2.internal

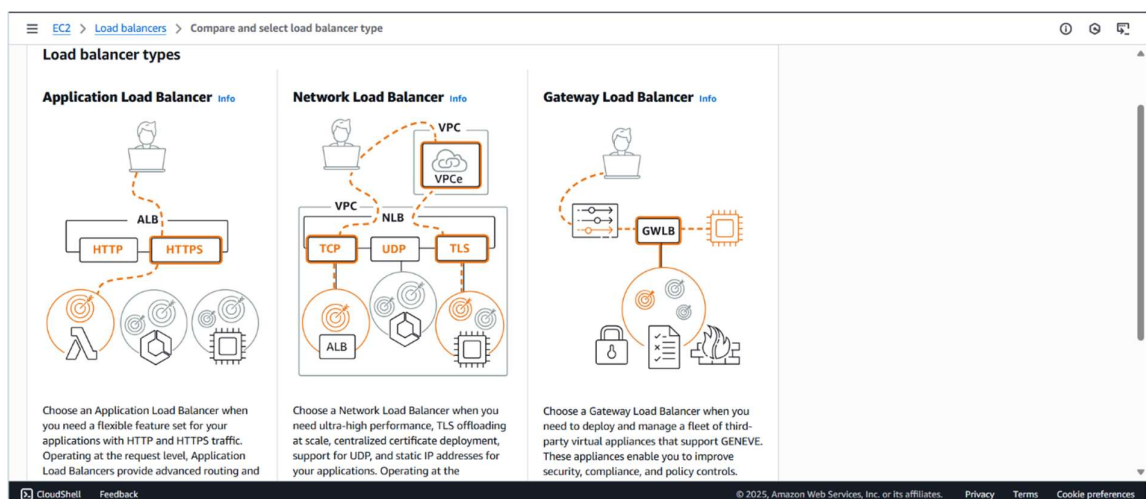
IP Address: fe80::a6:4dff:fe54:6e19%ens5

3. Configure a Network Load Balancer.

Purpose: A Network Load Balancer (NLB) is a high-performance AWS service designed to distribute TCP, UDP, and TLS traffic across multiple targets, such as Amazon EC2 instances, containers, and IP addresses, at the transport layer (Layer 4). Unlike the Application Load Balancer, the NLB focuses on ultra-low latency and high throughput, making it ideal for applications requiring extreme performance and scalability, such as gaming, VoIP, and real-time streaming.

Steps to Configure Network Load Balancer:

- Go to EC2 → Load Balancers → Create Load Balancer → Network Load Balancer.



The screenshot shows the 'Create Network Load Balancer' page in the AWS console. The 'Basic configuration' section is visible, containing the following fields and options:

- Load balancer name:** A text input field containing 'network-loadbalancer'. A note below states: 'Name must be unique within your AWS account and can't be changed after the load balancer is created. A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.'
- Scheme:** Two radio button options are shown: 'Internet-facing' (selected) and 'Internal'. The 'Internet-facing' option has a list of bullet points: 'Serves internet-facing traffic.', 'Has public IP addresses.', 'DNS name resolves to public IPs.', and 'Requires a public subnet.' The 'Internal' option has a list of bullet points: 'Serves internal traffic.', 'Has private IP addresses.', and 'DNS name resolves to private IPs.'
- Load balancer IP address type:** A section with a note: 'Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types.' It shows two radio button options: 'IPv4' (selected) and 'IPv6'.

Configure:

- Name: networkloadbanalcer
- Listeners: Add TCP listener (e.g., TCP 80)
- Availability Zones: Select AZs and subnets
- Optionally assign Elastic IPs for static routing

The screenshot shows the 'Create Network Load Balancer' page in the AWS Management Console. The breadcrumb navigation at the top reads 'EC2 > Load balancers > Create Network Load Balancer'. The page title is 'Create Network Load Balancer'. Below the title, there is a description: 'The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.' The 'VPC' section shows a dropdown menu with 'vpc-095ab90336dbc031d (test-vpc)' selected, with a 'Create VPC' button next to it. The 'Availability Zones and subnets' section has a checkbox for 'us-east-1a (use1-az1)' which is checked. Below it, a 'Subnet' dropdown shows 'subnet-0dac65a123f4f2c09' selected. The 'IPv4 address' section has a radio button for 'Assigned by AWS' which is selected, and a 'Use an Elastic IP address' button is disabled. Another checkbox for 'us-east-1b (use1-az2)' is checked, with a 'Subnet' dropdown showing 'subnet-0aa9ff8ce7f207219' selected. The footer of the console shows 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc.

- Go to EC2 → Target Groups → Create Target Group.
- Choose:
- Target type: instance or IP
- Protocol: TCP
- Port: e.g., 80
- VPC: Select your VPC

The screenshot shows the 'Create target group' page in the AWS Management Console. The breadcrumb navigation at the top reads 'EC2 > Target groups > Create target group'. The page title is 'Create target group'. Below the title, there is a description: 'Your load balancer routes requests to the targets in a target group and performs health checks on the targets.' The 'Basic configuration' section has a note: 'Settings in this section can't be changed after the target group is created.' The 'Choose a target type' section has three radio buttons: 'Instances' (selected), 'IP addresses', and 'Lambda function'. The 'Instances' option has a description: 'Supports load balancing to instances within a specific VPC. Facilitates the use of Amazon EC2 Auto Scaling to manage and scale your EC2 capacity.' The 'IP addresses' option has a description: 'Supports load balancing to VPC and on-premises resources. Facilitates routing to multiple IP addresses and network interfaces on the same instance. Offers flexibility with microservice based architectures, simplifying inter-application communication. Supports IPv6 targets, enabling end-to-end IPv6 communication, and IPv4-to-IPv6 NAT.' The 'Lambda function' option has a description: 'Facilitates routing to a single Lambda function. Accessible to Application Load Balancers only.' The footer of the console shows 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc.

- Configure health checks:
- Protocol: TCP or HTTP
- Port: traffic port or customs
- Register EC2 instances as targets.

EC2 > Target groups > Create target group

Step 1: Create target group
Step 2: Register targets

Register targets

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2/2)

Filter instances

☒	Instance ID	Name	State	Security groups	Zone
☒	i-02d71bf6c26e7a1e9	server2	Running	launch-wizard-5	us-east-1b
☒	i-02ae61d9bcf81de30	server1	Running	launch-wizard-6	us-east-1a

2 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

[Include as pending below](#)

EC2 > Target groups > network-target-group

Successfully created the target group: network-target-group. Anomaly detection is automatically applied to all registered targets. Results can be viewed in the Targets tab.

network-target-group

Details

arn:aws:elasticloadbalancing:us-east-1:085794372055:targetgroup/network-target-group/34ee62f6b24387b0

Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1	VPC vpc-095ab90336db031d
IP address type IPv4	Load balancer None associated		

2 Total targets	0 Healthy	0 Unhealthy	2 Unused	0 Initial	0 Draining
--------------------	--------------	----------------	-------------	--------------	---------------

0 Anomalous

Distribution of targets by Availability Zone (AZ)
Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets | Monitoring | Health checks | Attributes | Tags

EC2 > Load balancers > network-loadbalancer

Successfully created load balancer

network-loadbalancer

Details

Load balancer type Network	Status Provisioning	VPC vpc-095ab90336db031d	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone Z26RNL4JYFTOT1	Availability Zones subnet-0aa9ff8ce7f207219 us-east-1b (use1-az2) subnet-0dac65a123f4f2c09 us-east-1a (use1-az1)	Date created October 9, 2025, 15:42 (UTC+05:30)

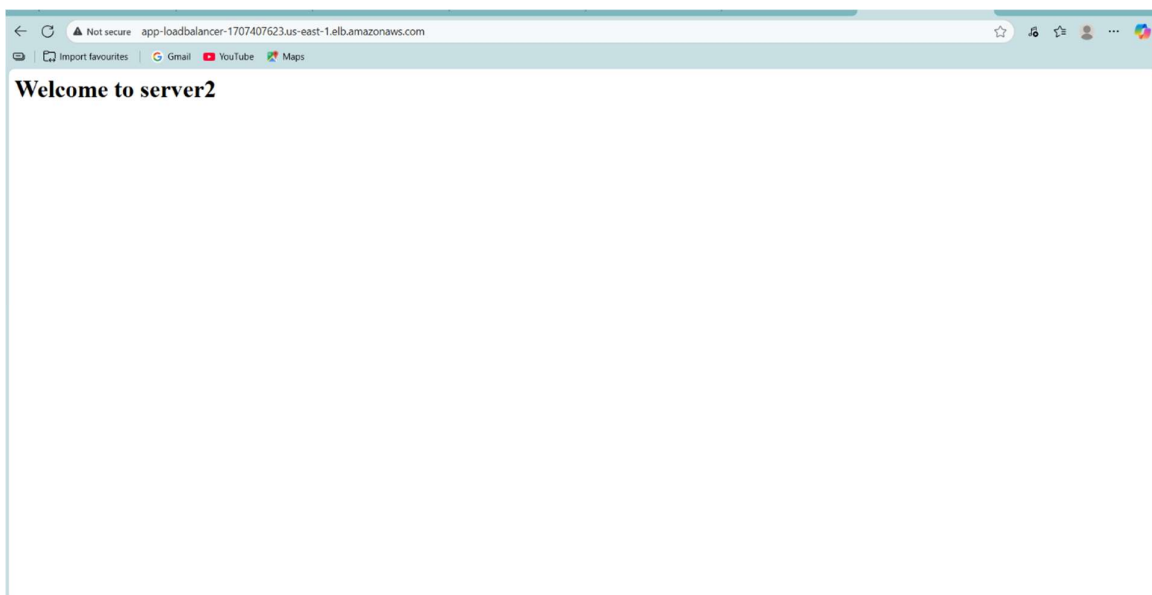
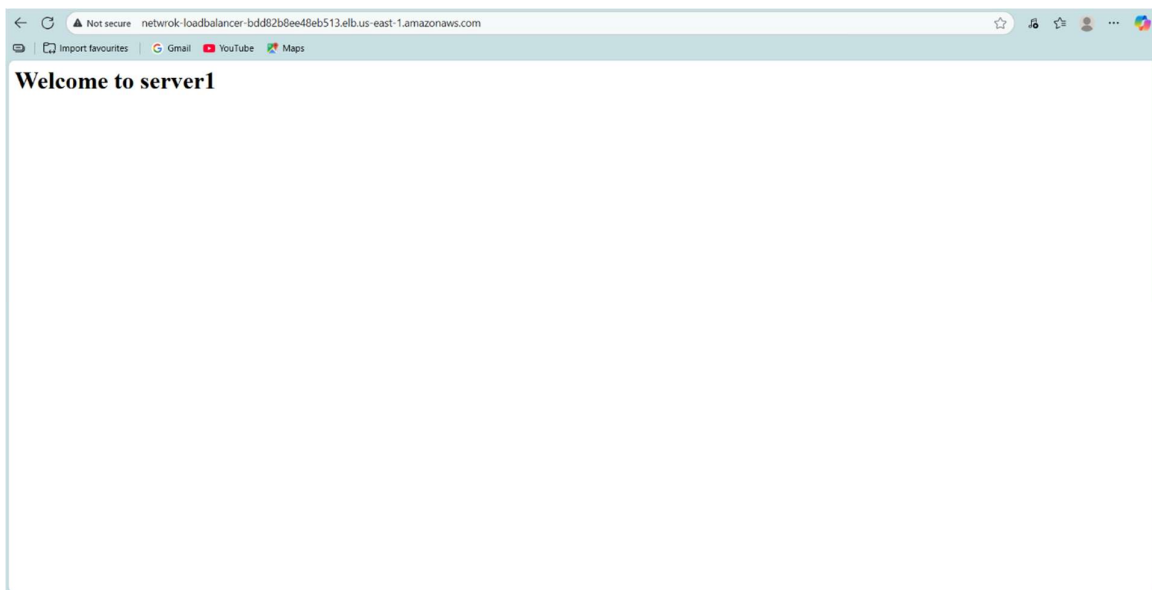
Load balancer ARN
[arn:aws:elasticloadbalancing:us-east-1:085794372055:loadbalancer/net/network-loadbalancer/bdd82b8ee48eb513](#)

DNS name info
[network-loadbalancer-bdd82b8ee48eb513.elb.us-east-1.amazonaws.com](#) (A Record)

Listeners | Network mapping | Resource map | Security | Monitoring | Integrations | Attributes | Capacity | Tags

Listeners (1)

[Add listener](#)

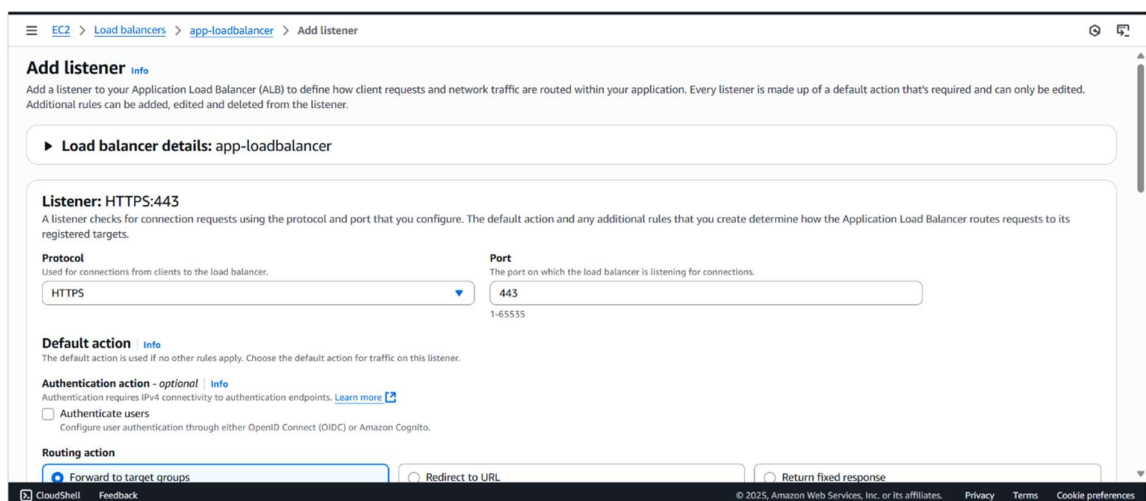


4. Attach SSL for the Application Load Balancer

Purpose: The primary purpose of this task is to encrypt all communication between clients and your application, preventing unauthorized access, data breaches, or tampering during transmission. By enabling HTTPS, you not only protect sensitive user information—such as login credentials, payment details, and personal data—but also build trust with your users, as modern browsers display a padlock icon to indicate a secure connection.

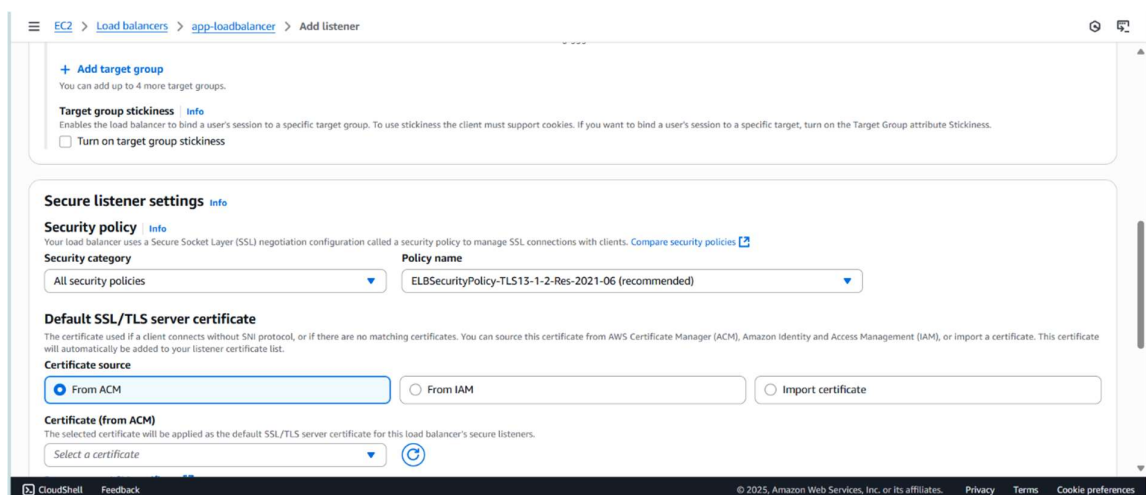
Steps to Attach SSL for the Application Load Balancer:

- Add HTTPS Listener to ALB
- Go to EC2 → Load Balancers → Your ALB → Listeners → Add Listener.
- Choose: Protocol: HTTPS, Port: 443



The screenshot shows the 'Add listener' page in the AWS Management Console. The breadcrumb trail is 'EC2 > Load balancers > app-loadbalancer > Add listener'. The page title is 'Add listener' with an 'info' link. Below the title is a description: 'Add a listener to your Application Load Balancer (ALB) to define how client requests and network traffic are routed within your application. Every listener is made up of a default action that's required and can only be edited. Additional rules can be added, edited and deleted from the listener.' A section titled 'Load balancer details: app-loadbalancer' is expanded. The 'Listener: HTTPS:443' section explains that the listener checks for connection requests using the protocol and port. It shows 'Protocol' set to 'HTTPS' and 'Port' set to '443'. The 'Default action' section states that the default action is used if no other rules apply. The 'Authentication action - optional' section has an 'info' link and a checkbox for 'Authenticate users'. The 'Routing action' section has three options: 'Forward to target groups' (selected), 'Redirect to URL', and 'Return fixed response'.

- Default action: Forward to your target group
- Attach SSL Certificate
- Under SSL certificate, select the ACM certificate for your domain.



The screenshot shows the 'Secure listener settings' page in the AWS Management Console. The breadcrumb trail is 'EC2 > Load balancers > app-loadbalancer > Add listener'. The page title is 'Secure listener settings' with an 'info' link. Below the title is a section for 'Security policy' with an 'info' link. It explains that the load balancer uses a Secure Socket Layer (SSL) negotiation configuration called a security policy. It shows 'Security category' set to 'All security policies' and 'Policy name' set to 'ELBSecurityPolicy-TLS13-1-2-Res-2021-06 (recommended)'. The 'Default SSL/TLS server certificate' section explains that the certificate is used if a client connects without the SNI protocol. It shows 'Certificate source' set to 'From ACM' (selected), with options for 'From IAM' and 'Import certificate'. The 'Certificate (from ACM)' section shows a dropdown for 'Select a certificate' and a refresh icon.

EC2 > Target groups > Create target group

Facilitates routing to a single Lambda function.

Accessible to Application Load Balancers only.

Application Load Balancer

Offers the flexibility for a Network Load Balancer to accept and route TCP requests within a specific VPC.

Facilitates using static IP addresses and PrivateLink with an Application Load Balancer.

Target group name

port443-target-group

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Protocol

Protocol for load balancer-to-target communication. Can't be modified after creation.

HTTPS

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

443

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

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EC2 > Load balancers > app-loadbalancer > Add listener

HTTPS

443

1-65535

Default action

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Authentication action - optional

Authentication requires IPv4 connectivity to authentication endpoints. [Learn more](#)

Authenticate users

Configure user authentication through either OpenID Connect (OIDC) or Amazon Cognito.

Routing action

Forward to target groups

Redirect to URL

Return fixed response

Forward to target group

Choose a target group and specify routing weight or [create target group](#)

Target group

port443-target-group

Target type: Instance, IPv4 | Target stickiness: Off

HTTPS

Weight

1

0-999

Percent

100%

+ Add target group

You can add up to 4 more target groups.

Target group stickiness

Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

Turn on target group stickiness

CloudShell

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AWS Certificate Manager > Certificates > 1a1a6c45-aeae-49ef-8d91-2accb41780c8

AWS Certificate Manager (ACM)

List certificates

Request certificate

Import certificate

AWS Private CA

1a1a6c45-aeae-49ef-8d91-2accb41780c8

Delete

Certificate status

Identifier

1a1a6c45-aeae-49ef-8d91-2accb41780c8

ARN

arn:aws:acm:us-east-1:085794372055:certificate/1a1a6c45-aeae-49ef-8d91-2accb41780c8

Type

Amazon Issued

Status

Pending validation

Info

Domains (0)

< 1 >

Domain

Status

Loading

Details

CloudShell

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- Go to ACM → Request a certificate.
- Choose public certificate.
- Enter your domain
- Validate via DNS (Route 53 auto-creates the record).

The screenshot shows the AWS Certificate Manager (ACM) console. On the left, there's a sidebar with the 'AWS Certificate Manager (ACM)' logo and links for 'List certificates', 'Request certificate', 'Import certificate', and 'AWS Private CA'. The main area is titled 'Certificates (8/8)' and contains a table of certificates. The table has columns for Certificate ID, Domain name, Type, Status, and In use. All certificates listed are for the domain 'www.razaq.live' and have a status of 'Issued'. There are 8 certificates in total, all of which are Amazon Issued and are currently in use.

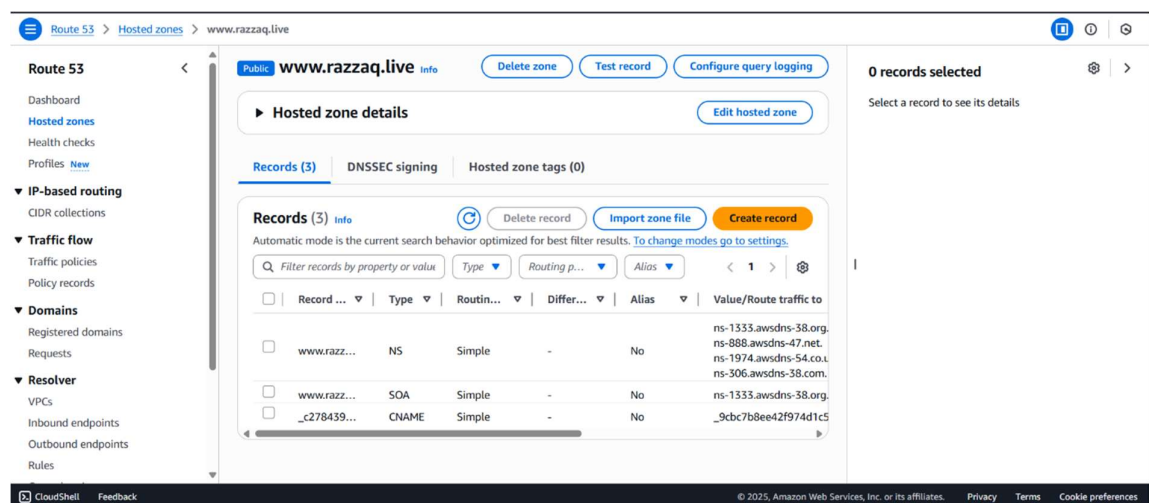
Certificate ID	Domain name	Type	Status	In use
9ca3b5a7-fb5b-4709-ad90-ea5e06d438c7	www.razaq.live	Amazon Issued	Issued	No
5b94246c-2d8e-4a31-b3f3-aea5013e890d	www.razaq.live	Amazon Issued	Issued	No
eb53b8bf-b208-4aef-a962-f5eeb020567a	www.razaq.live	Amazon Issued	Issued	No
24795e75-882c-4148-bf3a-e0a4d32f5635	www.razaq.live	Amazon Issued	Issued	No
eefbbcd7-7fc0-45d3-8540-49c932d8c688	www.razaq.live	Amazon Issued	Issued	No
8a3521aa-c69c-4ce4-b3ce-1166553d6d1f	www.razaq.live	Amazon Issued	Issued	No
1801a6c3-f1c3-4bda-8019-6ca7d0cc692	www.razaq.live	Amazon Issued	Issued	No
1a1a6c45-aeae-49ef-8d91-2accb41780c8	www.razaq.live	Amazon Issued	Issued	No

5. Map Application load balancer to R53.

Purpose: The purpose is to associate a human-readable domain name (e.g., `www.yourdomain.com`) with your ALB, making your application accessible to users via a custom URL instead of a complex IP address or AWS-generated DNS name. By mapping the ALB to Route 53, you ensure seamless traffic routing, improved reliability, and easier management of DNS records. This step is crucial for deploying production-ready applications, as it enhances user experience and simplifies access.

Steps to map the Application Load Balancer to R53:

- Go to EC2 → Load Balancers → Your ALB.
- Copy the DNS name
- Go to Route 53 Hosted Zone
- Navigate to Route 53 → Hosted Zones → YourDomain.com.



- Create Alias A Record
- Click Create Record.
- Set:
- Record name: leave blank for root domain
- Record type: A – IPv4 address
- Alias: Yes
- Alias target: Select your ALB from the dropdown (auto-populates if in the same region)
- Routing policy: Simple (or choose Weighted, Latency, etc.)

Route 53 > Hosted zones > www.razzaq.live > Create record

Create record info

Quick create record

Record 1

Record name info
 www.razzaq.live

Record type info
A – Routes traffic to an IPv4 address and some AWS resources

Alias ☒

Route traffic to info
Alias to Application and Classic Load Balancer
US East (N. Virginia)

Routing policy info
Simple routing

Evaluate target health ☒ Yes

[Add another record](#)

Switch to wizard

Delete

Keep blank to create a record for the root domain.

Alias hosted zone ID: Z35SXDOTRQ7X7K

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Route 53 > Hosted zones > www.razzaq.live

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles [New](#)
- IP-based routing
 - CIDR collections
- Traffic flow
 - Traffic policies
 - Policy records
- Domains
 - Registered domains
 - Requests
- Resolver
 - VPCs
 - Inbound endpoints
 - Outbound endpoints
 - Rules

Record for www.razzaq.live was successfully created.
Route 53 propagates your changes to all of the Route 53 authoritative DNS servers within 60 seconds. Use "View status" button to check propagation status.

[View status](#)

Records (4) DNSSEC signing Hosted zone tags (0)

Records (1/4) info [Delete record](#) [Import zone file](#) [Create record](#)

Automatic mode is the current search behavior optimized for best filter results. To change modes go to settings.

Filter records by property or value Type Routing p... Alias

Record ...	Type	Routin...	Differ...	Alias	Value/Route traffic to
☒ www.razz...	A	Simple	-	Yes	dualstack.app-loadbalancer-1707407623.us-east-1.elb.amazonaws.com
☐ www.razz...	NS	Simple	-	No	ns-1333.awsdns-38.org. ns-888.awsdns-47.net. ns-1974.awsdns-54.co.l ns-306.awsdns-38.com.
☐ www.razz...	SOA	Simple	-	No	ns-1333.awsdns-38.org.
☐ _c278439...	CNAME	Simple	-	No	_9cbc7b8ee42f974d1c5

Record details

[Edit record](#)

Record name
www.razzaq.live

Record type
A

Value
dualstack.app-loadbalancer-1707407623.us-east-1.elb.amazonaws.com.

Alias
Yes

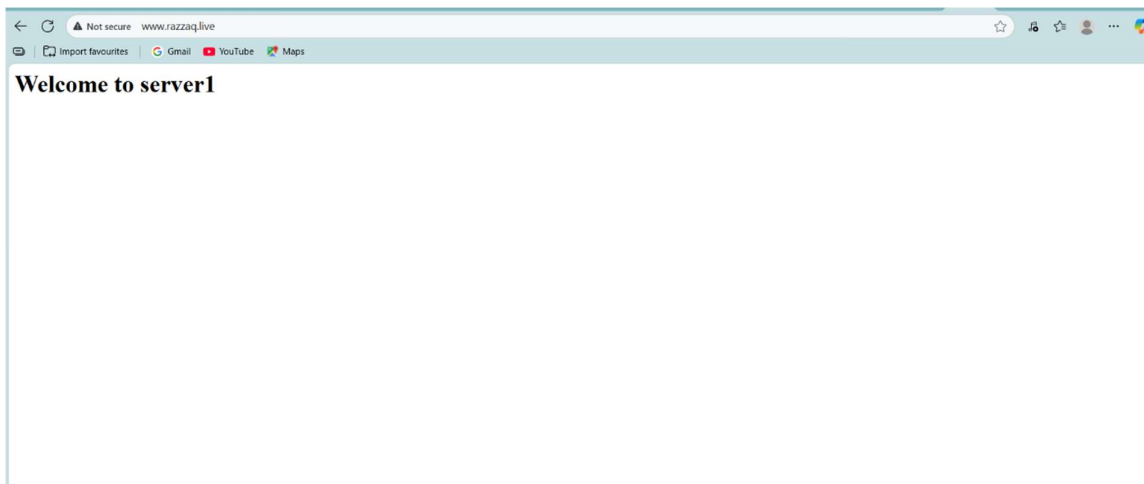
TTL (seconds)
-

Routing policy
Simple

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- Visit www.razzaq.live.
- You should be routed to your ALB and see your app (e.g., NGINX welcome page).

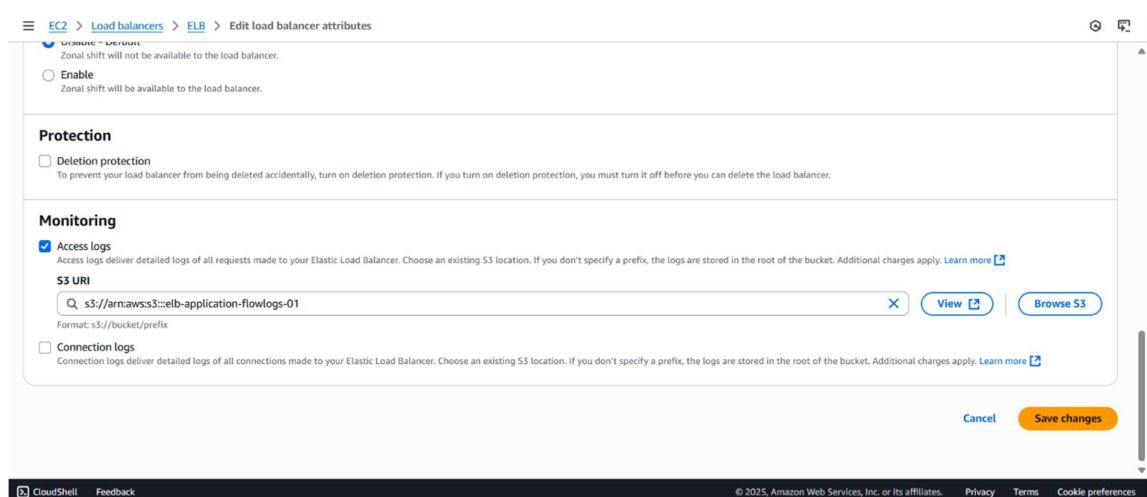
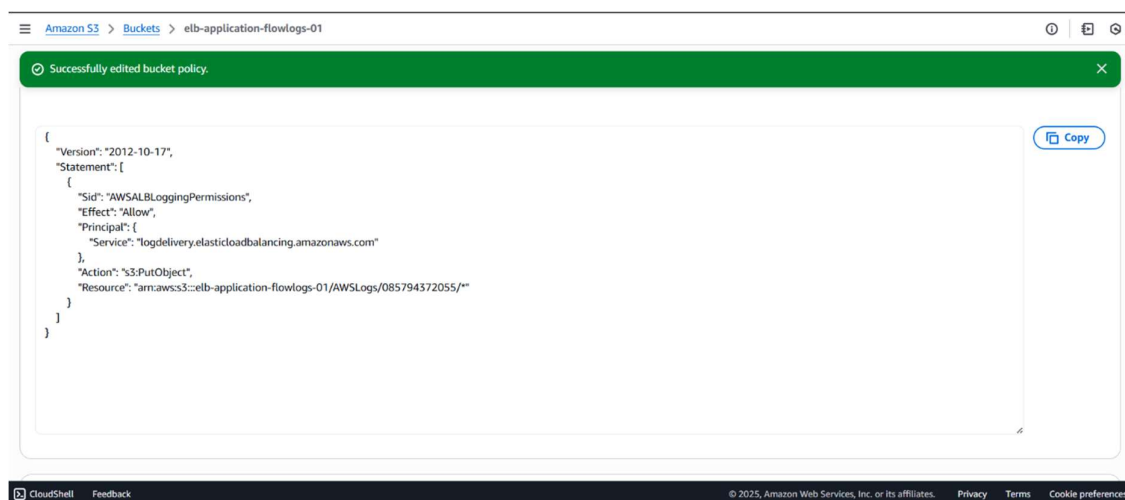


6. Push the application load balancer logs to S3.

Purpose: This task involves configuring your Application Load Balancer (ALB) to store access logs in an Amazon S3 bucket. The purpose is to capture detailed records of all incoming requests, including client IP addresses, request paths, latency, and response codes. By pushing ALB logs to S3, you enable long-term storage, analysis, and monitoring of traffic patterns, security threats, and performance bottlenecks. This is essential for auditing, troubleshooting, and compliance, ensuring your application remains secure and optimized.

Steps to Push ALB logs to S3:

- Select an already created bucket
- Set Bucket Policy for ALB Logs.
Go to the bucket → Permissions → Bucket Policy → Add



- Go to EC2 Dashboard → Load Balancers → Select your ALB
 - Description tab → “Edit attributes.”
 - Enable Access logs

